

Pentium Processor Architecture & Superscalar Operation — MPCA Module 5

Definition: The **Intel Pentium Processor** is a 32-bit superscalar CISC microprocessor capable of executing **two instructions per clock cycle** using two parallel 5-stage integer pipelines (**U-Pipe** and **V-Pipe**), an integrated on-chip Floating-Point Unit (FPU), and **Dynamic Branch Prediction Logic**.

1. Detailed Technical Explanation

1. Pentium Architecture Block Diagram

PENTIUM INTERNAL ARCHITECTURE

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| 64-bit External Data Bus <====> 32-bit Address Bus (4 GB Address Space) |
| [ 8 KB Code Cache (I-Cache) ]      [ 8 KB Data Cache (D-Cache) ]      |
| [ Branch Target Buffer (BTB) ] -----> 2-bit Dynamic Branch Prediction |
| [ Instruction Decode Unit ]                                               |
| / \                                                                       |
| [ U-Pipeline (5-Stage) ] [ V-Pipeline (5-Stage) ] [ Pipelined 80-bit FPU ] |
| (Executes any instruction) (Executes simple pairable) (8-stage math unit) |
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```

2. Superscalar Integer Pipelines (U and V Pipes)

The Pentium can issue two integer instructions simultaneously in parallel if they meet **Instruction Pairing Rules**:

- U-Pipeline:** Can execute any instruction from the x86 instruction set.
- V-Pipeline:** Executes simple integer instructions (e.g., **MOV**, **ADD**, **SUB**, **INC**, **DEC**, **CMP**, **JMP**, **CALL**).

The 5 Pipeline Stages:

- PF (Prefetch)** : Fetches instructions from 8 KB Code Cache into prefetch buffers.
- D1 (Decode-1)** : Decodes instruction opcode, determines pairability between U & V.
- D2 (Decode-2)** : Computes Effective Memory Address (Displacement + Base + Index).
- EX (Execute)** : ALU executes integer operations; accesses Data Cache if needed.
- WB (Writeback)** : Updates destination registers and status flags.

3. Dynamic Branch Prediction & Branch Target Buffer (BTB)

- Branch Target Buffer (BTB):** An on-chip cache storing branch instructions, target jump addresses, and branch history bits.

- **2-bit Branch History State Machine:**

- **00** : Strongly Not-Taken
 - **01** : Weakly Not-Taken
 - **10** : Weakly Taken
 - **11** : Strongly Taken
 - Achieves over **90% branch prediction accuracy**, eliminating branch penalty stalls from pipeline flushing.
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4. Must-Write Points for Exams

- Pentium has a **64-bit external data bus** and **32-bit address bus**.
 - It features separate Harvard-style **8 KB Code Cache** and **8 KB Data Cache** on-chip.
 - Dual-issue U and V pipelines achieve execution throughput of up to 2 instructions per clock cycle (**IPC = 2**).
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5. Quick Recall Flow

Pentium -> Superscalar -> U & V Dual Pipelines (5-stages: PF, D1, D2, EX, WB) -> 8KB Code/Data Caches -> BTB 2-bit Branch Prediction